Department of Mathematics and Statistics
Texas Tech University

Exam II

Fall 2026
Practice Exam

Name (print): _____ **R-Number:** _____ **Seat Number:** _____

- Please read the directions for all problems carefully.
- Show all solutions in the space provided. If necessary, the back of a page may be used to complete an answer.
- Show a sufficient and reasonable amount of work to justify your answers.
- Please circle your final answer.
- Use of any electronic device during the examination is strictly prohibited. Please keep smartphones, smartwatches, and calculators out of view.

My signature below demonstrates that I will comply with the directions above, the work on this exam is my own, and I did not receive or give assistance on the exam.

Signature: _____

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1. (10 pts) Consider the function $f(x) = 2x^3 - 5$.

(a) (6 pts) Use the definition of the derivative to show that $f'(x) = 6x^2$.

(b) (4 pts) Use part (a) to find $f'(-2)$ and $f''(x)$.

2. (10 pts) Use the definition of the derivative to show that $f'(x) = -\sin(x)$ if $f(x) = \cos(x)$.

Recall:

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta, \qquad \sin(\alpha + \beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha.$$

You may use $\lim_{h\to 0} \dfrac{\sin h}{h} = 1$ and $\lim_{h\to 0} \dfrac{\cos h - 1}{h} = 0$.

3. (30 pts) Determine the derivative of the following functions. (3 pts each)

(a) $f(x) = 4x^3 - 6x + 11$

(b) $g(x) = \sqrt[3]{x^5}$

(c) $h(x) = \dfrac{x^4 + 3x - 2}{x^3}$

(d) $f(x) = x^6 \ln(x)$ (Do not simplify.)

(e) $y = \dfrac{\sin(x)}{x^3}$ (Do not simplify.)

(f) $s(t) = \dfrac{1}{\sin(t)}$

(g) $f(x) = e^{(x^3 - 4x + 5)}$

(h) $r(\theta) = \big[\ln(\cos\theta)\big]^3$

(i) $f(x) = \ln\big(\sqrt{x^8 + x^3 + 5}\big)$

(j) $y = \sqrt{\dfrac{x^3 + 2}{x^2 - 7}}$ (Do not simplify.)

4. (10 pts) Consider the curve defined implicitly by $x^2y + y^3 = 10$.

(a) (6 pts) Use implicit differentiation to find $\dfrac{dy}{dx}$.

(b) (4 pts) Verify that the point $(1, 2)$ lies on the curve, then evaluate $\dfrac{dy}{dx}$ at $(1, 2)$.

Total: _____ / 60